Knowledge Management with (Bio)Croissant*

Steffen Vogler :: BioCypher Workshop :: 15 June 2026

* … *BioCroissant* is the name of the working group, *Croissant* is the name of the exchange format

Data is Source Code for AI

- **Croissant makes datasets machine-readable without changing the data**
  - One metadata layer across storage locations
  - Compatible for Agentic tools

- **It describes everything AI needs to know about data**
  - Access, structure, licensing, provenance, annotation, and intended use

Croissant: A Metadata Format for ML-Ready Datasets

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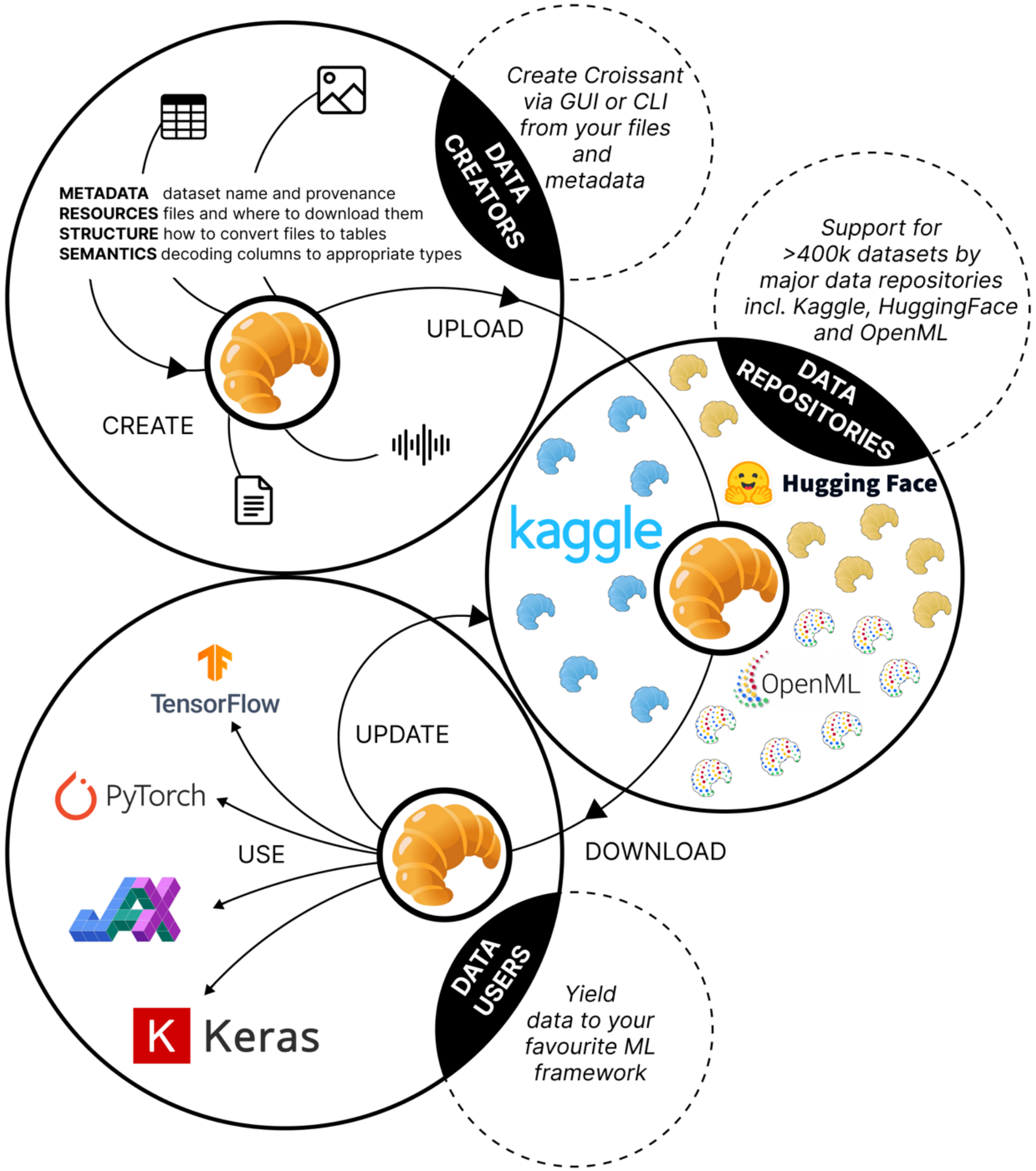


Data is Source Code for AI

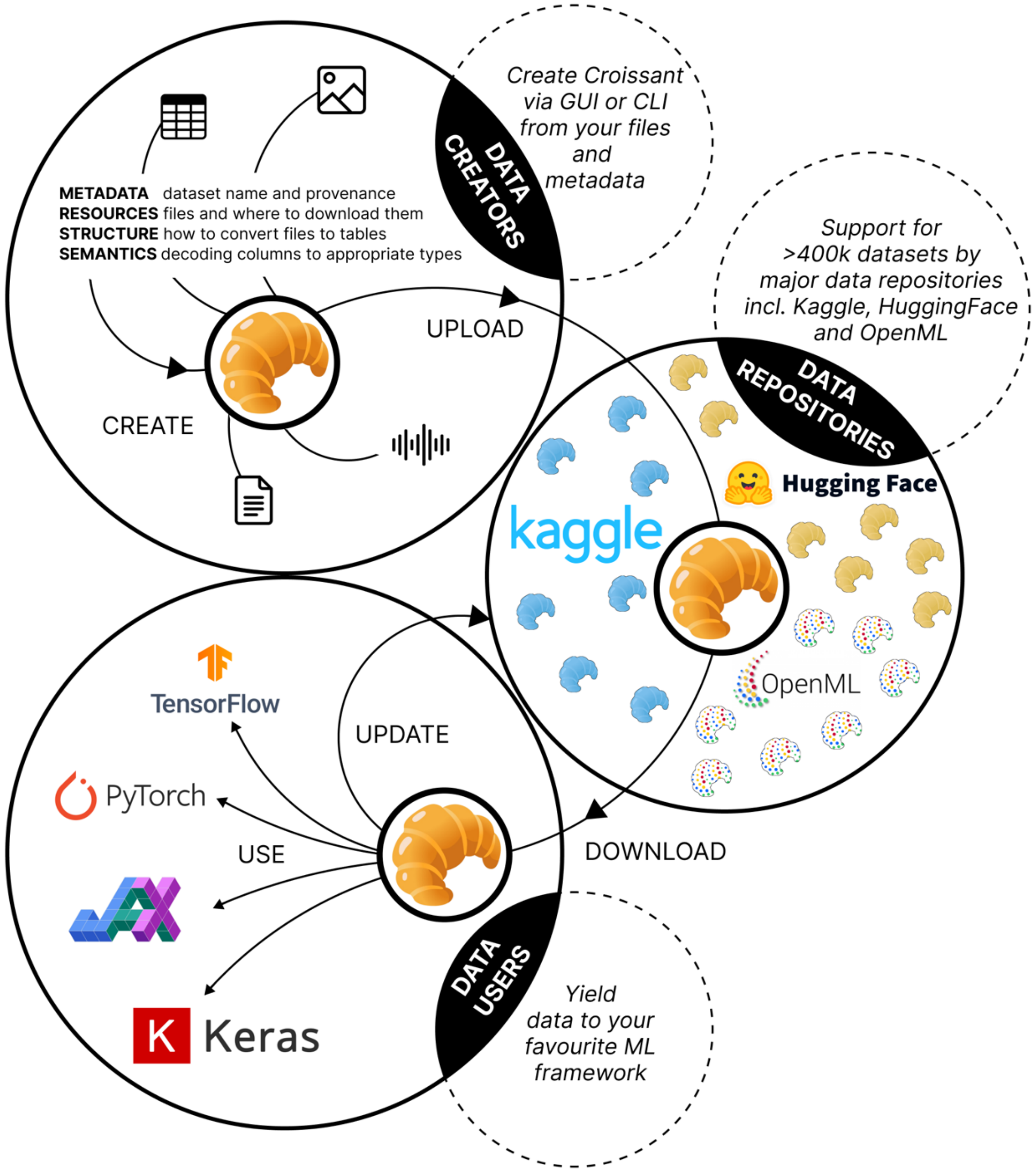
- Croissant vocabulary does not require changing the underlying data representation

- Unified view over different layouts and formats

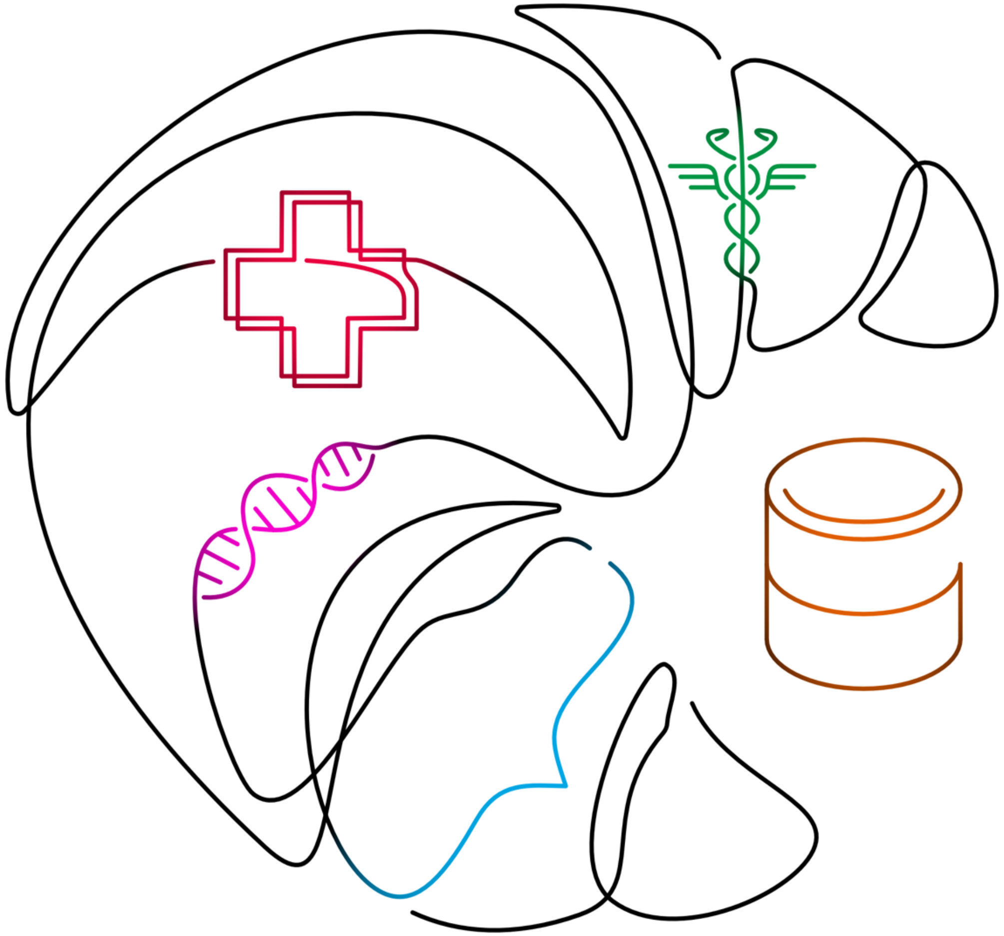
- Croissant adds structure and semantic
  - where is it stored? how to download?
  - how to cite? kind of license
  - how to read it? how to prepare the data?
  - who has annotated? annotation software used?

- Croissant file (json-type) follows schema.org syntax
  - Web-compatible
  - Machine readable & Searchable
  - Apache Beam support

- Responsible AI extension
  - Data Cards with harmonized vocabulary



Croissant on the inside



```
{
  "@type": "sc:Dataset",
  "name": "minimal_example_with_recommended_fields",
  "description": "This is a minimal example, including the required and the recommended fields.",
  "license": "https://creativecommons.org/licenses/by/4.0/",
  "url": "https://example.com/dataset/recipes/minimal-recommended",
  "distribution": [
    {
      "@type": "cr:FileObject",
      "@id": "minimal.csv",
      "name": "minimal.csv",
      "contentUrl": "data/minimal.csv",
      "encodingFormat": "text/csv",
      "sha256":
"48a7c257f3c90b2a3e529ddd2cca8f4f1bd8e49ed244ef53927649504ac55354"
    }
  ],
  "recordSet": [
    {
      "@type": "cr:RecordSet",
      "name": "examples",
      "description": "Records extracted from the example table, with their schema.",
      "field": [
        {
          "@type": "cr:Field",
          "name": "name",
          "description": "The first column contains the name.",
          "dataType": "sc:Text",
          "references": {
            "fileObject": { "@id": "minimal.csv" },
            "extract": {
              "column": "name"
            }
          }
        },
        {
          "@type": "cr:Field",
          "name": "age",
          "description": "The second column contains the age.",
          "dataType": "sc:Integer",
          "references": {
            "fileObject": { "@id": "minimal.csv" },
            "extract": {
              "column": "age"
            }
          }
        }
      ]
    }
  ]
```

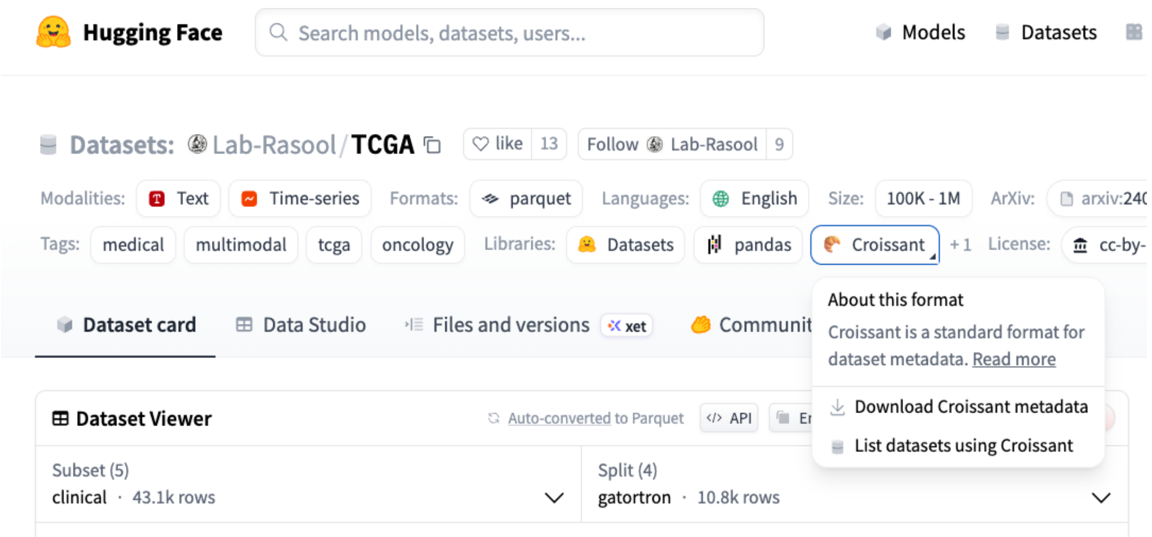
Metadata

Structure

Resources

Semantics

How to use it?




```
pip install mlcroissant
```

```python
# 1. Point to a local or remote Croissant file
import mlcroissant as mlc
url = "https://huggingface.co/api/datasets/fashion_mnist/croissant"
# 2. Inspect metadata
print(mlc.Dataset(url).metadata.to_json())
# 3. Use Croissant dataset in your ML workload
import tensorflow_datasets as tfds
builder = tfds.core.dataset_builders.CroissantBuilder(
    jsonld=url,
    record_set_ids=["record_set_fashion_mnist"],
    file_format='array_record',
)
builder.download_and_prepare()
# 4. Split for training/testing
train, test = builder.as_data_source(split=['default[:80%]', 'default[80%:]'])
```

For practitioners – easy to use, just load your data from a single file (w/ context)

Why BioCroissant WG is needed – what AI for Health is missing

In analogy to the European Healthcare Data Space (EHDS):

Governance above; technical execution below

EHDS Governance Layer

Legal framework · purpose limitation · access rights

HealthDCAT-AP

High level · Not auto-loadable

Dataset Discovery & Approval

Catalogues · portals · HDABs · access requests

HealthDCAT-AP

Findable · Not executable

Technical Dataset Specification

Features · splits · schemas · provenance · task definitions

BioCroissant

Machine-readable · Auto-loadable

Secure Processing & Analysis

SPEs · TREs · federated ML · AI model training

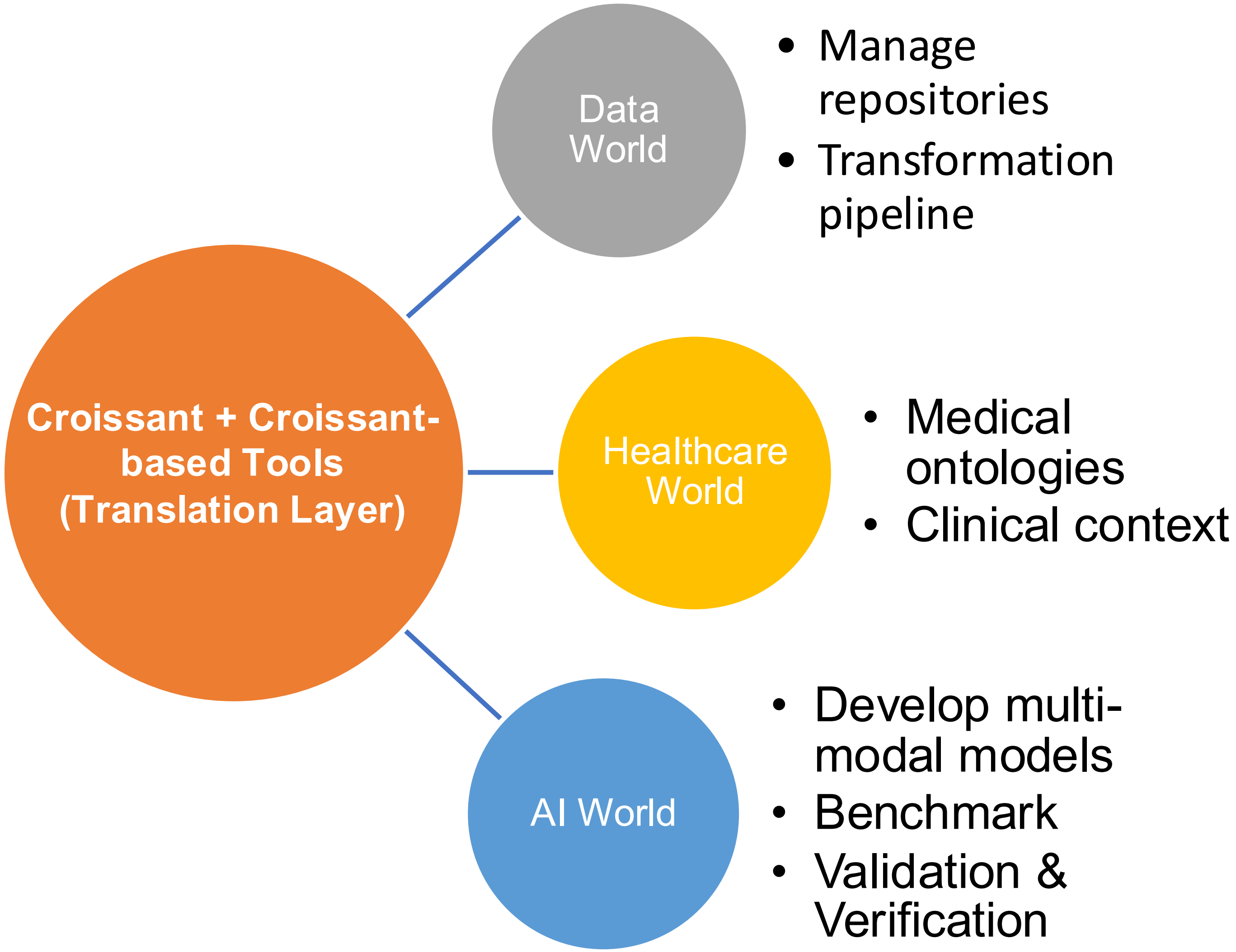
Execution layer

Requires EHDS-compliant access grant

- Non-normative & voluntary
- Downstream of EHDS governance
- Enables auto-loading of datasets
- EHDS-compatible technical profile

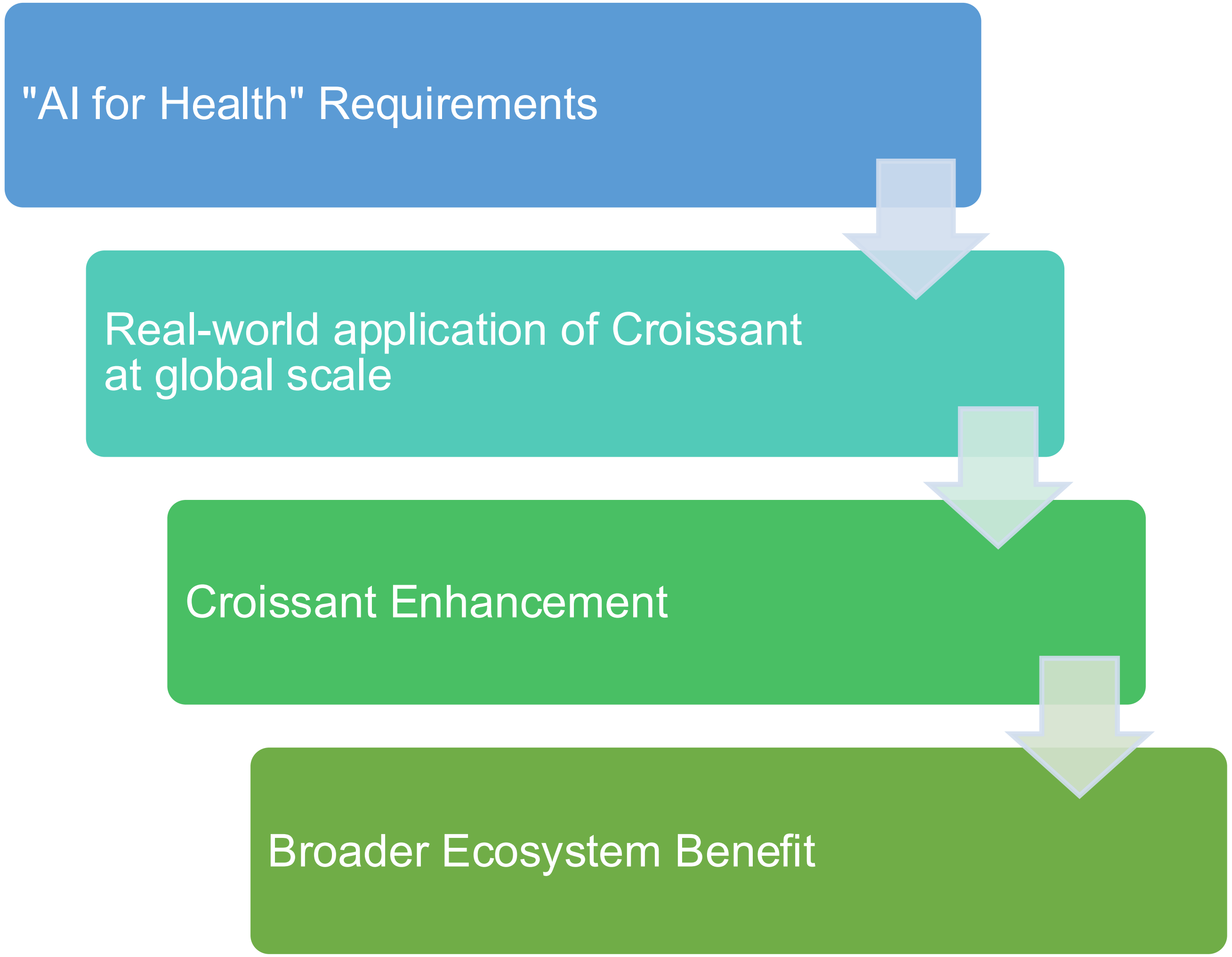
Goal: Bridging Three Communities with one data language

- Smart **metadata wrapping** across modalities, formats, and systems
- **Complexity stays under the hood.**
  Tooling handles structure, provenance, licensing, and semantics
- **Experts stay in their lane.**
  Each community works in familiar tools and concepts.



Selected to Build a Global Medical Data Catalogue

- Croissant has been selected by **ITU/WHO Global Initiative on AI for Health**
- "AI for Health" brings **real-world requirements** to Croissant
- Global benchmarking turns **theory into practice**
- **Feedback** flows back into Croissant and **improves** it



Building a Powerful Tool Ecosystem

BioCroissant Repository

mlcommons / BioCroissant

Code | Issues | Pull requests | Agents | Discussions

BioCroissant Public

🥐 BioCroissant — Metadata Sharing for (Bio)Medical AI

BioCroissant is a sub-working group of the Croissant initiative with strong ties to the MLCommons Medical working group that is developing benchmarks and best practices. BioCroissant focuses on exploring and extending the Croissant standard for the biomedical domain, strengthening the technological link between **AI** and **biomedical communities**.

Its goal is to make medical and life-science data standards work seamlessly with the AI ecosystem—across ML frameworks, platforms, and Responsible AI tooling.

Built with real-world **clinical**, **research**, and **regulatory** needs in mind, BioCroissant bridges the gap between **ML engineers**, **healthcare professionals**, and **policy makers** working to build trustworthy health data ecosystems (e.g. academic health data consortia or international initiatives such as the European Health Data Space, EHDS).

🚀 Why BioCroissant?

Modern biomedical AI depends on high-quality, interoperable data—yet many datasets today remain siloed, inconsistently documented, or difficult to reuse.

BioCroissant provides:

- **A simple, structured metadata schema** for biomedical datasets
- **Built-in FAIR alignment** (Findable, Accessible, Interoperable, Reusable)
- **Compatibility with national and cross-border health data initiatives**
- **Seamless integration with ML workflows** through clean, machine-readable metadata
- **Governance-friendly transparency**, supporting reproducibility, auditability, and clinical trust

Whether you are building models, conducting clinical research, or shaping data policy, BioCroissant helps ensure your data is **ready for science**, **ready for AI**, and **ready for regulation**.

https://github.com/mlcommons/BioCroissant

- Call for Sharing/Contribution/Feedback
- Roadmap is driven my discussion in the WG
- My favorite next topic:
  - Biomedical imaging formats
- FHIR support is around the corner

Discussions

Votes	Discussion	Comments
↑ 2	**Dual-compatibility Croissant files following domain ontologies (e.g. OMOP, FHIR, and BioSchema profiles)** steffenvogler started on Apr 28 in Ideas	2
↑ 1	**Data modalities beyond imaging** charliejharrison started 3 weeks ago in General	2
↑ 5	**Support for biomedical imaging formats (WSI, microscopy, multi-resolution images, OCT etc)** steffenvogler started on Apr 28 in Ideas	11
↑ 2	**Representing longitudinal patient-centric data** steffenvogler started on Apr 28 in Ideas	0

Croissant Baker

Croissant Baker

Run Tests passing | python 3.10+ | uv | License MIT | pypi v0.3.0 | PRs welcome

Automatically generate Croissant JSON-LD metadata for ML datasets — e.g. for PhysioNet, NeurIPS Datasets & Benchmarks submissions, or any platform that benefits from standardized dataset metadata.

📖 Documentation

Installation

```
pip install croissant-baker
```

or with uv:

```
uv add croissant-baker
```

https://github.com/MIT-LCP/croissant-baker

- CLI to auto-generate metadata files for local data
- Plays nicely with coding agents
  - Deterministic instead of guessing fields

Croissant Baker: Metadata Generation for Discoverable, Governable, and Reusable ML Datasets

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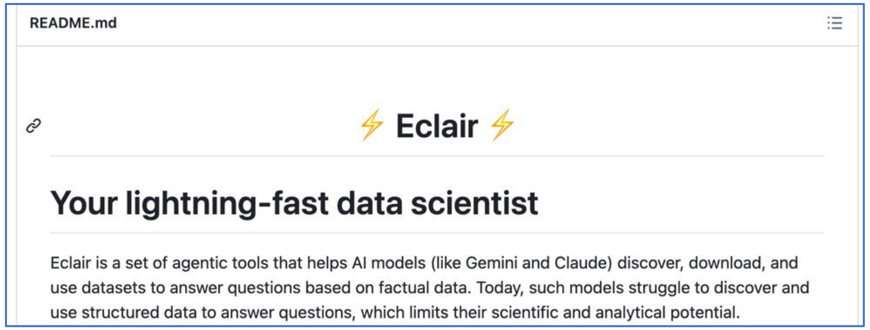
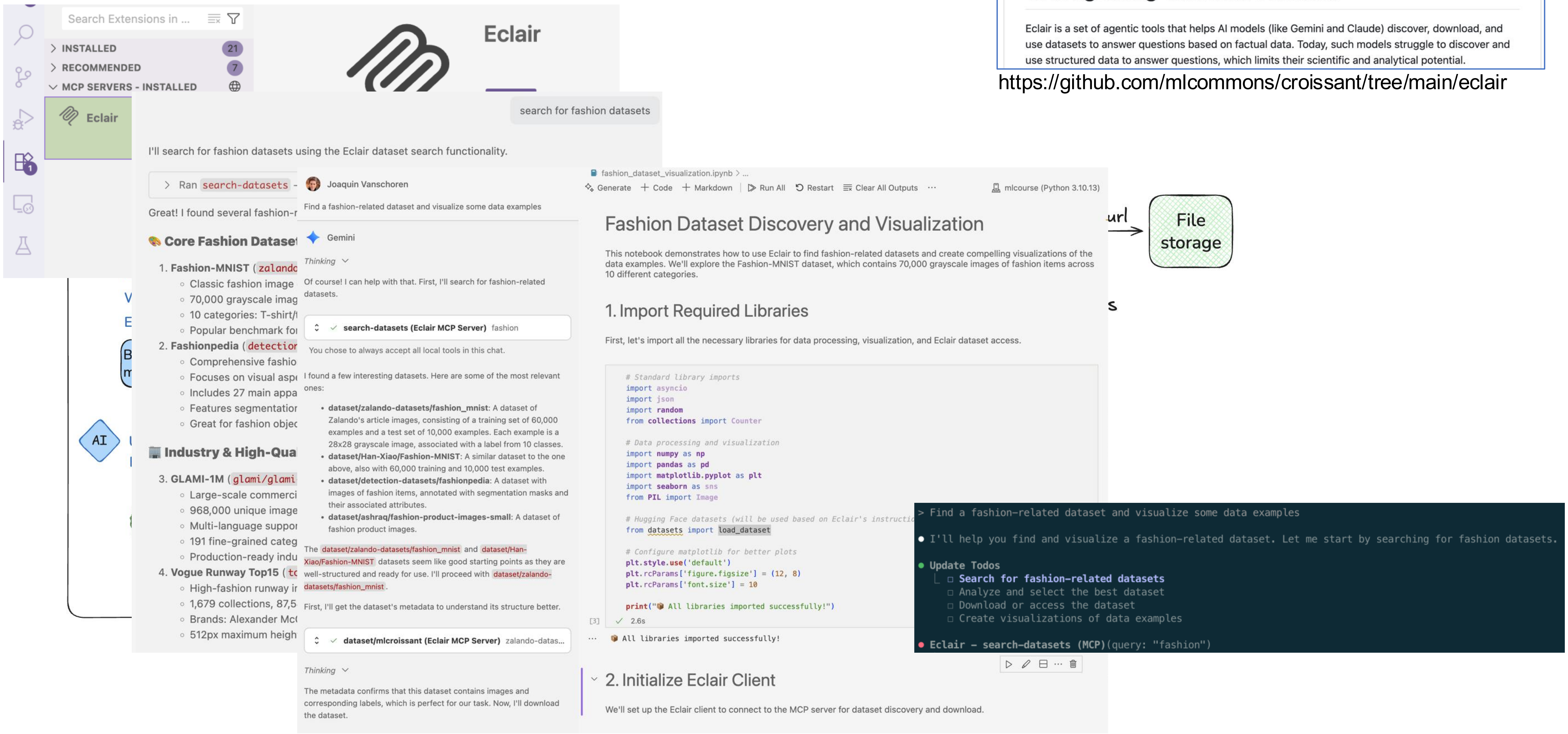
https://github.com/MIT-LCP/croissant-baker
https://pypi.org/project/croissant-baker/
https://lcp.mit.edu/croissant-baker/

- Manuscript has strong medical focus
- Submitted to NeurIPS “Evaluations and Datasets”
- https://arxiv.org/abs/2605.15079

Eclair – Croissant’s MCP server solution



https://github.com/mlcommons/croissant/tree/main/eclair



Call to action: Join the Movement!

Tool Developers

Integrate Croissant support into data repositories, analysis tools, and ML frameworks to expand ecosystem

Policy Makers

Reference Croissant standards in data governance frameworks to promote interoperability and FAIR compliance

Researchers

Contribute to BioCroissant working group, share implementation experiences, and help refine health-specific extensions

Data Providers

Adopt Croissant for health datasets to improve discoverability and enable responsible reuse across platforms